

CR-004: Student Input Field Validation for Add and Edit Operations

1. Change Request Details

Field	Details
Change Request ID	CR-004
Title	Student Input Field Validation for Add and Edit Operations
Priority	High
Difficulty	Low
Change Type	Frontend and Backend Validation Enhancement
Affected Version	Student Management System v1.0 baseline
Dependency	Existing Student Add/Edit functionality and API endpoints

2. Business Requirement

Administrators currently can enter student information without sufficient validation when adding or editing students. For example, a phone number containing only a few digits may be accepted, and names with fewer than three characters or weak passwords may also be entered.

As incorrect or insecure information can affect data quality and account security, the system should validate all student input fields before accepting the data.

The system should ensure that student names, registration number, email address, phone number, and password follow the required validation rules.

3. Current Behaviors

- StudentFormModal.jsx performs only basic required-field checking.
- A phone number with fewer than 10 digits may be accepted.
- A phone number with more than 10 digits may be entered.
- Non-numeric characters may be entered into the phone number field.
- First name and last name do not have a minimum 3-character requirement.
- Invalid characters may be entered into name fields.
- Email format is not sufficiently validated.
- Registration number does not have comprehensive input validation.
- New student passwords do not enforce a strong password policy.
- Backend validation does not comprehensively validate all student input fields.
- Invalid data may potentially be submitted directly to the backend without passing client-side validation.

4. Requested Change

Add comprehensive validation to the student Add and Edit operations.

The implementation must:

- Validate all required student fields before submission.
- Require a valid registration number.
- Require first name and last name to contain at least 3 characters.
- Allow valid alphabetic characters in first name and last name.
- Reject invalid numbers or inappropriate special characters in names.
- Validate the email address using a valid email format.
- Require the phone number to contain exactly 10 digits.
- Allow only numeric characters in the phone number.
- Reject phone numbers containing fewer than 10 digits.
- Reject phone numbers containing more than 10 digits.
- Require a password when creating a new student.
- Require the password to contain at least 8 characters.
- Require at least one uppercase letter in the password.
- Require at least one lowercase letter in the password.
- Require at least one number in the password.
- Require at least one special character in the password.
- Display clear validation error messages to administrators.
- Prevent invalid data from being submitted from the frontend.
- Apply server-side validation to prevent client-side validation bypass.
- Preserve existing Add, Edit, and Delete functionality.
- Maintain existing administrator authentication and authorization.
- Continue securely hashing passwords before storing them in the database.

5. Scope

In Scope

- Student Add form validation.
- Student Edit form validation.
- Registration number validation.
- First name validation.
- Last name validation.
- Minimum 3-character name validation.
- Name character validation.
- Email format validation.
- Exactly 10-digit phone validation.

- Numeric-only phone validation.
- Strong password validation for new students.
- Client-side validation.
- Server-side validation.
- Clear validation error messages.
- Prevention of invalid form submission.
- Prevention of invalid data from being stored.
- Preservation of existing CRUD functionality.
- Preservation of existing authentication and authorization.

Out of Scope

- Database schema redesign.
- New student management API endpoints.
- Changes to administrator authentication.
- Changes to administrator authorization.
- Changes to student search functionality.
- Changes to student Delete functionality.
- Pagination.
- Changes to unrelated modules.
- Changing the existing password hashing mechanism.

6. Expected / Potentially Affected Files

- `frontend/src/components/StudentFormModal.jsx`
- `backend/controllers/studentController.js`
- `backend/models/studentModel.js` if required for supporting validation-related database constraints
- Existing student form stylesheet if required

Students may modify additional files if required. Any additional files must be identified and justified in the Implementation Report.

7. Functional & Security Requirements

- **FR-001:** All required student fields shall be validated before submission.
- **FR-002:** Registration number shall not be empty.
- **FR-003:** First name shall contain at least 3 characters.
- **FR-004:** Last name shall contain at least 3 characters.
- **FR-005:** First name and last name shall contain valid alphabetic characters.
- **FR-006:** Email shall follow a valid email format.
- **FR-007:** Phone number shall be required.
- **FR-008:** Phone number shall contain exactly 10 digits.
- **FR-009:** Phone number shall contain numeric characters only.
- **FR-010:** Phone numbers containing fewer than 10 digits shall be rejected.
- **FR-011:** Phone numbers containing more than 10 digits shall be rejected.
- **FR-012:** Password shall be required when creating a new student.
- **FR-013:** Password shall contain at least 8 characters.
- **FR-014:** Password shall contain at least one uppercase and one lowercase letter.
- **FR-015:** Password shall contain at least one number.
- **FR-016:** Password shall contain at least one special character.
- **FR-017:** Invalid input shall prevent form submission.
- **FR-018:** Clear validation error messages shall be displayed.
- **FR-019:** Validation shall apply to both Add and Edit operations.
- **FR-020:** Valid student information shall be successfully saved.
- **SEC-001:** Existing administrator authentication shall remain unchanged.
- **SEC-002:** Existing administrator authorization shall remain unchanged.
- **SEC-003:** Server-side validation shall prevent client-side validation bypass.
- **SEC-004:** Invalid student information shall not be stored in the database.
- **SEC-005:** Passwords shall not be stored as plain text.
- **SEC-006:** Passwords shall continue to be securely hashed using the existing password hashing mechanism.
- **SEC-007:** Validation errors shall not expose sensitive information.
- **SEC-008:** Existing Edit and Delete authorization shall remain unchanged.

8. Acceptance Criteria

The CR is successfully implemented when:

- **AC-001:** Administrators cannot submit the form when required fields are empty.
- **AC-002:** First names containing fewer than 3 characters are rejected.
- **AC-003:** Last names containing fewer than 3 characters are rejected.
- **AC-004:** Names containing invalid numbers or inappropriate special characters are rejected.

- **AC-005:** Invalid email addresses are rejected.
- **AC-006:** Phone numbers containing fewer than 10 digits are rejected.
- **AC-007:** Phone numbers containing more than 10 digits are rejected.
- **AC-008:** Phone numbers containing non-numeric characters are rejected.
- **AC-009:** A valid 10-digit phone number is accepted.
- **AC-010:** Weak passwords are rejected when creating a new student.
- **AC-011:** A password containing at least 8 characters, uppercase, lowercase, number, and special character is accepted.
- **AC-012:** A valid student can be successfully created.
- **AC-013:** Invalid student information cannot be successfully created.
- **AC-014:** Invalid student information cannot be successfully updated.
- **AC-015:** A valid student can be successfully updated.
- **AC-016:** Direct API requests containing invalid data are rejected by the backend.
- **AC-017:** Existing Edit and Delete functionality continues to affect the correct student.
- **AC-018:** Existing administrator authentication and authorization continue to work correctly.

9. Expected Outcome

Administrators can add and edit students using properly validated information, while invalid or insecure data is rejected before it is stored.

The system will enforce minimum name length, valid name characters, valid email format, exactly 10-digit phone numbers, and strong passwords for new student accounts.

Both client-side and server-side validation will help maintain data quality and prevent validation bypass, while existing API, database, authentication, authorization, and CRUD functionality remain unchanged.

10. Implementation Report Requirement

After implementation, the Final Exam Software Change Implementation Report shall include:

- Implementation summary.
- Affected files.
- Validation rules implemented for each student field.
- Strong password validation implementation.
- Important technical/security changes.
- Client-side validation test cases and results.
- Server-side validation test cases and results.

- Acceptance criteria verification.
- Screenshots/evidence showing validation errors.
- Screenshots/evidence showing valid student creation.
- Screenshots/evidence showing valid student update.
- Evidence showing backend rejection of invalid requests.
- Evidence that existing Edit and Delete functionality remains functional.
- Final status and outstanding issues.